Activity 5: Normality Detection Using Q-Q Plots

Course: DSA0601 – Data Handling and Visualization

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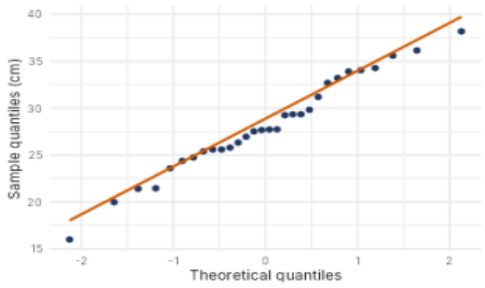
R. No.: 36

DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

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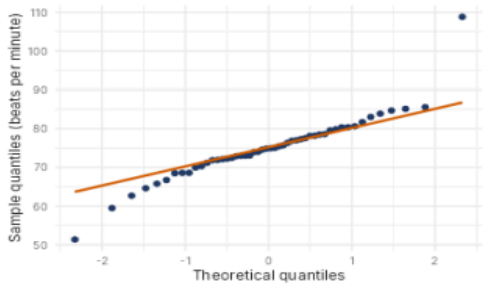
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: plant height (cm), n = 30



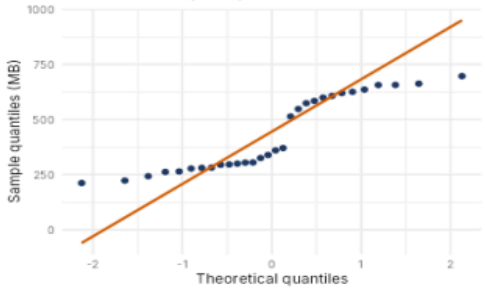
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: resting pulse (beats per minute), n = 50



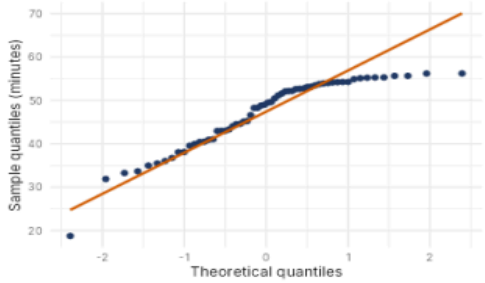
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: mobile data used per day (MB), n = 30



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: commute time (minutes), n = 60



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 2 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Part A

Chart 1: Plant height (cm), n = 30

- Normal? Yes
- Shape: Normal
- Evidence in the plot: The points lie approximately along the straight reference line, with only small deviations.
-

Chart 2: Resting pulse (beats per minute), n = 50

- Normal? No
- Shape: Heavy-tailed
- Evidence in the plot: The points at both ends deviate from the reference line, with extreme low and high values.
-

Chart 3: Mobile data used per day (MB), n = 30

- Normal? No
- Shape: Right-skewed
- Evidence in the plot: The points show a strong curved pattern and deviate from the straight reference line, especially in the upper quantiles.
-

Chart 4: Commute time (minutes), n = 60

- Normal? No
- Shape: Left-skewed
- Evidence in the plot: The points bend away from the reference line, with a lower-tail deviation and a flattening upper tail.

Part B

B1. Chart 2 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods.

The departure is heavy tails with extreme values. I would check for data-entry errors and outliers. If the values are valid, I would consider robust methods such as the median and IQR, or a suitable transformation if appropriate.

Code

```
library(ggplot2)

df <- data.frame(value = c(
  12, 15, 18, 20, 22, 24, 25, 26,
  27, 28, 29, 30, 31, 32, 33, 35,
  36, 38, 40, 42
))

ggplot(df, aes(sample = value)) +
  stat_qq() +
  stat_qq_line()
```

Output

